Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 12

Union, Intersection, Difference, and Complement

Covers Chartrand & Zhang, Section 2.2, pp. 60–68

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Translate to logic, then compute

The dictionary

$x \in A \cap B$	$(x \in A) \wedge (x \in B)$
$x \in A \cup B$	$(x \in A) \vee (x \in B)$ — *inclusive*
$x \in \overline{A}$	$\sim(x \in A)$
$x \in A - B$	$(x \in A) \wedge \sim(x \in B)$
$x \in A \oplus B$	$(x \in A) \oplus (x \in B)$

Any question about sets can be turned into a question about statements, and you have four weeks of practice with those.

Three mistakes to expect

1. **Reading $\cup$ as exclusive.** An element in both sets is still in the union.
2. **Assuming $A - B = B - A$.** There is no commutative law for difference.
3. **Forgetting the universal set.** $\overline{A}$ is meaningless until U is named.

Part II: Practice Problems

Section A — Computing

1. Let $U = \{1, 2, \ldots, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$, and $C = \{2, 4, 6, 8\}$. Compute each.

(a) $A \cap B$

(b) $A \cup B$

(c) $A - B$

(d) $B - A$

(e) $A \oplus B$

(f) $\overline{A}$

(g) $\overline{A \cup B}$

(h) $A \cap B \cap C$

2. With the same U, A, B as above, verify by direct computation that $A \cap B \subseteq A \cup B$, and explain why this holds for *every* pair of sets.

Section B — Membership tables

3. Build a membership table for $A - B$ and for $A \cap \overline{B}$, and state whether they are equal.

$x \in A$	$x \in B$	$x \in \overline{B}$	$x \in A - B$	$x \in A \cap \overline{B}$
T	T			
T	F			
F	T			
F	F			

Section C — Reasoning

4. Answer each with a brief justification.

(a) If $A \subseteq B$, what is $A \cap B$? What is $A \cup B$?

(b) If A and B are disjoint, what is $A \oplus B$?

(c) Can $A - B = B - A$ ever hold? Under what condition?

Section D — Writing

5. In five to eight complete sentences, explain why a Venn diagram is useful for *understanding* a set identity but does not *prove* it. Say what does constitute a proof, and mention what a membership table covers that a picture does not.

Section E — Challenge (optional)

6. Two sets carve the universe into four regions, and three sets into eight. How many regions do n sets create, and what does that have to do with membership tables?